

Bitcoin's Quantum Divide

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1. Introduction

Since the first block was mined on January 3rd, 2009, the viability of Bitcoin as a legitimate monetary asset has rested upon the assumption that there exist certain problems that are easy to verify but extremely computationally challenging to solve. So far, this assumption has held for the fundamental problem that is central to the field of Elliptic Curve Cryptography. Formally known as the Elliptic Curve Discrete Logarithm Problem (ECDLP), this problem stipulates that, given a specific point on an elliptic curve that was generated by multiplying a secret scalar value by a known generator point, there is no algorithm to efficiently reveal that scalar value. Though this might seem like abstract theory, this problem lies at the heart of the Bitcoin network. To put it into concrete terms, these seemingly arbitrary points and scalars directly map to Bitcoin's key structure: A Bitcoin user's private key is this secret scalar value, and their public key is the point that was generated by multiplying that private key by a generator point. Given the computational intractability of solving the ECDLP, it is impossible to derive that private key from the public key. This mathematical certainty allows Bitcoin users to trust that the private keys they use to unlock funds and sign transactions cannot be re-derived and stolen by an attacker, so long as they are generated and stored securely. It has been understood for some time, however, that this security model only holds within the classical models of computation under which the ECDLP's intractability was first established. As our understanding of subatomic physics and material sciences matures, a cutting-edge field of computer science now threatens to upend how computers process information. By leveraging the strange behaviors of subatomic particles, quantum computers can offer exotic solutions to problems that are computationally intractable on classical machines. Unfortunately for the Bitcoin network, the ECDLP is one of these problems.

To make matters worse, across much of the world, the adoption of quantum-resistant cryptography (more commonly called post-quantum cryptography, or PQC) is being directed by a mixture of both public and private standards bodies, many of which offer different recommendations and mandates about which post-quantum digital signing algorithms are acceptable to use. Some countries directly adopt the United States' guidelines published by the National Institute of Standards and Technology (NIST). Others follow more loosely, recommending most of NIST's algorithms, but choosing to specify key sizes and recommend additional, non-NIST approved schemes. Some countries, citing industrial leverage and security concerns, prefer not to follow U.S. standards at all, instead opting to develop domestic cryptographic suites. As of March 2026, there are discussions underway about how the Bitcoin community could sidestep this problem with an algorithm-agnostic Bitcoin Improvement Proposal (BIP) that is currently under review¹. However, even if a BIP successfully

¹Beast, Hunter, Ethan Heilman, and Isabel Foxen Duke. "BIP 360: Pay-to-Merkle-Root (P2MR)." BIP 360: Pay-to-Merkle-Root (P2MR), December 18, 2024. <https://bip360.org/bip360.html>.

implements a soft fork that lets users pick their favorite PQC signature scheme, it doesn't necessarily eliminate the transitional friction caused by this disagreement over standards. Since validating a coin's history requires the validator to use the signature verification software corresponding to the scheme used to generate each signature, users might be unable to personally validate the spend history of the bitcoin that they receive, thus diminishing confidence in the network.

The problem of a post-quantum migration, then, is not exclusively cryptographic. Even though the Bitcoin protocol might be designed to function as a neutral, globally decentralized system of money, it is bound by physical computing and routing resources that are subject to the political regulations and restrictions of the jurisdictions in which they reside. Selecting only the signature schemes developed and promoted by a single jurisdiction or geopolitical bloc could create the appearance that the network implicitly endorses that political power or coalition as a technological authority. This could alienate bitcoin users and holders that live in regions whose cryptographic preferences are omitted from the supposedly neutral architecture of this global currency. Trying to mitigate this disagreement by stuffing various mathematically different signature schemes into the protocol would fundamentally alter how Bitcoin's network handles signatures and could catalyze pushback from custodial institutions that are bound by regulatory and auditing frameworks. This paper asks how cryptographic standards bodies and geopolitical rivalries might shape which quantum-resistant solutions are written into the Bitcoin network, and what the consequences for the global adoption of bitcoin might be if a cross-jurisdictional consensus fails to emerge. Through a comparative analysis of Bitcoin's successful 2021 Taproot upgrade and Bitcoin's impending post-quantum migration, I will argue that the latter is likely to encounter political fragmentation because the institutions that increasingly drive Bitcoin's global adoption are not necessarily free to embrace whichever cryptographic solution the community selects.

In the following pages, I will begin by summarizing the current state of quantum computing and Bitcoin's post-quantum transition, including developments surrounding BIP 360, and the active BTQ Technologies testnet. I will then trace the upgrade process of the 2021 Taproot upgrade to establish a baseline understanding of how protocol upgrades are accepted across Bitcoin's decentralized network, and to identify the critical preconditions that allowed Taproot to succeed. Following this examination, I will analyze whether the impending post-quantum transition shares these same preconditions in order to predict how it might diverge from Taproot's model of success, especially focusing on the consequences of geopolitical fragmentation and contradictory guidance from standards bodies. Finally, we will use these conclusions to speculate about possible outcomes of this post-quantum transition, and how they might affect the long-term viability of bitcoin as a global currency.

2. The Quantum Threat to Bitcoin

Quantum computers are no longer a theoretical technology; they exist at companies, universities, and research institutions all around the world. The reason we don't consider them a threat at present is because they currently operate in what researchers call the "NISQ" (Noisy Intermediate-Scale Quantum) Era. This means that our best quantum computers only have a small to intermediate number of qubits available to perform computations, and that the qubits they do have are highly error prone. Interference from heat, electromagnetic noise, and even other subatomic particles can cause these qubits to lose their quantum state, thus preventing them from performing the complex, sustained calculations necessary to threaten modern cryptography.² Looking beyond the NISQ era, we can envision a cryptographically relevant quantum computer (CRQC), which is the name given to an imagined quantum computer that does have the computational power and resilience to actually break modern encryption. For a quantum computer to become a CRQC, it needs two things: qubits and fault tolerance. Depending on the type of quantum computer, adding qubits can be achieved by building increasingly complex chips using superconductors like aluminum or niobium or by trapping atoms via electromagnetic fields or laser beams.³ Improving fault tolerance, meaning the ability to correct errors caused by qubit interference, is often achieved using logical qubits, which are conceptual qubits that have their state contained across multiple physical qubits as a form of redundancy.⁴ While CRQC's do not yet exist, some research suggests we are beginning to accelerate towards a world where they do. According to a 2025 report put together by the Global Risk Institute and evolutionQ, a collection of 26 global quantum computing experts assessed that there is a 28-49% probability of a CRQC existing within the next 10 years, and a 51-70% probability of one existing within the next 15 years.⁵ Google, which is arguably the leading institution in the field of quantum research and development, announced in March 2026 that they believe quantum frontiers "may be closer than they appear," and that they are updating their timeline for post-quantum migration to 2029.⁶

²Dargan, James. "What Is NISQ Quantum Computing?" The Quantum Insider, September 24, 2025. <https://thequantuminsider.com/2023/03/13/what-is-nisq-quantum-computing/>.

³Wright, Katherine. "What's a Qubit? 3 Ways Scientists Build Quantum Computers." Scientific American, May 28, 2025. <https://www.scientificamerican.com/article/whats-a-qubit-3-ways-scientists-build-quantum-computers/>.

⁴"Logical Qubit." RSS. Accessed March 18, 2026. <https://www.quera.com/glossary/logical-qubit>.

⁵Mosca, Michele, and Marco Piani. Rep. *QUANTUM THREAT TIMELINE REPORT 2025*. Toronto, Ontario: Global Risk Institute, 2026.

⁶Adkins, Heather. "Quantum Frontiers May Be Closer than They Appear." Google, March 25, 2026. <https://blog.google/innovation-and-ai/technology/safety-security/cryptography-migration-timeline/>.

2.1 Bitcoin's Quantum Attack Vectors

In assessing the threat CRQC's pose to the Bitcoin network, one must consider the two primary attack vectors through which a quantum adversary could violate the network's integrity. The first involves extracting the public keys that are embedded in the locking and unlocking scripts attached to bitcoin transactions. Since the blockchain was designed to be transparent and immutable, any bitcoin address that has had its raw public key included in a past transaction's script is vulnerable to what is known as a "long exposure attack." A long exposure attack is a quantum attack in which an adversary has the luxury of a long time-frame available to derive an address's private key from its public counterpart because that public key has already been written into the ledger, which cannot be retroactively altered to obfuscate those keys. This form of exposure was standard practice for some of the oldest and most widely used output types. In Pay-to-Public-Key (P2PK), for example, the recipient's public key was included directly in the locking script, making it permanently visible once the transaction was confirmed. Spending those funds requires only a valid signature corresponding to that key. Later output types, like Pay-to-Public-Key-Hash (P2PKH), improved on this design by including only a hash of the public key in the locking script. However, when the recipient spends those funds, they must reveal both the public key and a corresponding signature in the unlocking script, thereby exposing the key.⁷ Long exposure attacks are particularly dangerous because any wallet that has had its public key written into the ledger is permanently at risk unless users migrate these funds to a quantum-safe output, or a protocol change makes them altogether unspendable.

Short exposure attacks, unlike their counterparts, do not have the luxury of being exploitable over an indefinite period. They similarly extract public keys to derive the corresponding private key, but they do not target public keys that are already written into the blockchain. Instead, a short exposure attack occurs when an attacker sets up a node so that they can pluck public keys from out of the mempool.⁸ All transactions, regardless of the output type must include a public key as part of the spending proof so that the sender's transaction signature can be verified before the transaction is written into a block. For output types that are vulnerable to long-exposure attacks (P2PK, P2MS, P2TR), the public key is directly exposed on the blockchain for use in its validation. For the output types that obfuscate the public key on the blockchain (P2PKH, P2WPKH, P2SH, P2WSH, P2MR), a public key must be included in the transaction broadcast so that the nodes can validate the transaction before accepting it into the mempool. If a quantum attacker sets up a node and is watching the incoming transactions, they could grab a public key, derive its private key, and then write a new

⁷Beast, Hunter, Ethan Heilman, and Isabel Foxen Duke. "BIP 360: Pay-to-Merkle-Root (P2MR)." BIP 360: Pay-to-Merkle-Root (P2MR), December 18, 2024. <https://bip360.org/bip360.html>.

⁸Short for "*Memory Pool*," the mempool is the temporary buffer where unconfirmed transactions wait after being validated by nodes and before being selected by a miner to be written into a block. Each node has its own mempool but also propagates valid transactions across the peer-to-peer node network to make transactions widely visible for review and acceptance by miners across the world.

transaction sending that spender's money to their own wallet address. For this transaction to be confirmed, the attacker would then need to do one of two things: either construct the next block, replacing the original transaction with their competing one, or broadcast the fraudulent transaction with a higher fee than the original to incentivize another miner to confirm it in the next block for them. In both cases, since a new block is mined roughly every 10 minutes, the attacker only has a small window to pick a public key, derive its private key, make a new transaction, and then replace the original transaction in the next block. While a short exposure attack would take a very powerful CRQC to execute, it is still a serious vulnerability because the Bitcoin network does not yet have an output type that could mitigate this attack. Unlike long exposure attacks, the remediation of short exposure vulnerabilities will almost certainly require at least one post-quantum signature scheme.⁹

2.2 Post-Quantum Cryptography

Cryptographic algorithms are classified into families based on the type of math they use. When it comes to the research and development of cryptography that is resistant to quantum-based attacks (often called post-quantum cryptography, or PQC), the two primary families are Hash-based algorithms, and Lattice-based algorithms. Bitcoin developers and researchers are actively evaluating a small number of leading candidate algorithms in both families. Under lattice-based algorithms, the primary signature scheme being examined is ML-DSA. ML-DSA stands for "Module Lattice Digital Signing Algorithm" and is the primary signature recommendation of the United States' National Institute of Standards and Technology (NIST) standards body.¹⁰ The primary downside to ML-DSA is that the signatures are, at minimum, 38 times larger than Schnorr signatures, which the Bitcoin network currently uses. FN-DSA is another lattice-based NIST selection, though an FN-DSA standard hasn't been formalized as of the writing of this paper. FN-DSA produces much smaller signatures but is more complex and difficult to implement securely than ML-DSA.¹¹ Under hash-based algorithms, we have SLH-DSA. This signature scheme is also NIST-approved, and it functions based on the well-studied and tested properties of existing hashing functions like SHA-256. Signatures are large and signing is slow, but this scheme has a longer history of resisting attacks than the newer lattice-based algorithms. There are other hash-based signature schemes being studied, too, such as XMSS and LMS. These are very similar to SLH-DSA, though not approved for general use by NIST because small errors in implementation can be catastrophic for their security.¹² These

⁹Ibid.

¹⁰Ivezic, Marin. "Inside Nist's PQC: Kyber, Dilithium, and SPHINCS+." PostQuantum, September 24, 2025. <https://postquantum.com/post-quantum/nists-pqc-technical/#crystals-kyber-lattice-based-key-encapsulation-mechanism>.

¹¹Ibid.

¹²Ishikiriya, Shizuka. "LMS versus XMSS versus SLH-DSA." wolfSSL, August 14, 2025. <https://www.wolfssl.com/lms-versus-xmss-versus-slh-dsa/>.

five schemes are not the only post-quantum algorithms currently available; Russia and China, for example, are both developing domestic PQC algorithm suites that have yet to be released.¹³ These five are, however, the most likely candidates for Bitcoin's migration.

2.3 The Current State of Bitcoin's Post-Quantum Migration

As of February 2026, the Bitcoin Network has taken its first steps towards migrating the network towards quantum-safe protocols through the merging of BIP 360 with the official "bitcoin improvement proposals" (BIPs) GitHub repository. This proposal defines a new output type, called "Pay-to-Merkle-Root" (P2MR), which would create a new output type that allows a sender to send bitcoin to recipient by locking the bitcoin to a Merkle root of the recipient's unlocking scripts. This system remediates long exposure attacks because the only information written to the blockchain is a Merkle root (which is just a SHA-256 hash). The recipient only reveals their public key at spend-time, for the 10-minute window necessary for the nodes to validate the recipient's locking script and Merkle root, and then add the transaction to a block.¹⁴ Since this current iteration of BIP 360 isn't intended to address short-exposure attacks, it does not attempt to select and incorporate a post-quantum signature scheme into the protocol. It does, however, make possible the incorporation of more than one post-quantum signature scheme because P2MR spends are ultimately validated through the spender's script leaves, which could generate and check a signature for any algorithm supported by the Bitcoin scripting language opcodes. Beyond BIP 360, teams of academic and corporate researchers are working to design, test, and share their own post-quantum solutions. One such effort is a post-quantum fork of the Bitcoin network developed by BTQ Technologies, a Canadian quantum technologies company that specializes in quantum-resistant infrastructure.¹⁵ This project is not intended to replace the actual Bitcoin network. Rather, it is a testnet that swaps Schnorr signatures for ML-DSA, incorporates BIP-360's P2MR output type by adding five new ML-DSA opcodes, and bumps up the block size to 64 MiB to account for larger signatures.¹⁶ The testnet code is open-source so that developers, miners, node operators and bitcoin users can easily test this specific post-quantum cryptographic architecture and search for incompatibilities

¹³Ivezic, Marin. "Sovereignty in the PQC Era: Standards, Trust, and Crypto-Agility." PostQuantum, February 2, 2026. <https://postquantum.com/post-quantum/sovereignty-quantum-pqc/#russia-developing-national-quantum-resistant-algorithms>.

¹⁴Beast, Hunter, Ethan Heilman, and Isabel Foxen Duke. "BIP 360: Pay-to-Merkle-Root (P2MR)." BIP 360: Pay-to-Merkle-Root (P2MR), December 18, 2024. <https://bip360.org/bip360.html>.

¹⁵Corp., BTQ Technologies. "BTQ Technologies Launches Bitcoin Quantum Testnet, Marking 17 Years since Bitcoin's Genesis Block with First Quantum-Safe Fork." PR Newswire: press release distribution, targeting, monitoring and marketing, 2026. <https://www.prnewswire.com/news-releases/btq-technologies-launches-bitcoin-quantum-testnet-marking-17-years-since-bitcoins-genesis-block-with-first-quantum-safe-fork-302658425.html>.

¹⁶Quantum, Bitcoin. "Bitcoin Quantum FAQ." FAQ | Bitcoin Quantum. Accessed April 10, 2026. <https://bitcoinquantum.com/faq>.

and vulnerabilities. In March 2026, Blockstream, the prominent bitcoin development firm founded by Adam Back, successfully deployed post-quantum signature verification on the Liquid sidechain using the Simplicity smart contract language and SHRINCS, a bitcoin-tailored variant of SLH-DSA with smaller signatures.¹⁷ These examples are just a few of the many promising efforts currently underway to turn theorized post-quantum applied cryptography into real-world deployments.

3. Bitcoin Governance and the Taproot Upgrade

On November 14, 2021, at block height 709,632, a new era of the Bitcoin network began. Block 709,632 was the moment at which the “Taproot” protocol upgrade was activated. This upgrade was first proposed in 2018 by Bitcoin developer Greg Maxwell, and comprised three BIPs written by Pieter Wuille, Tim Ruffing, A.J. Townes and Jonas Nick.¹⁸ Broadly speaking, the Taproot upgrade made the Bitcoin network more scalable, programmable and private by replacing ECDSA (the core cryptography utilized by the software) with Schnorr signatures, which are a slightly different Elliptic-Curve signature scheme. For an analysis of how standards bodies and geopolitical rivalries could affect Bitcoin's post-quantum migration, the technical research and development that produced Taproot's code are not particularly relevant. What is important, however, is understanding how Taproot generated enough of a consensus to achieve activation. To do this, we must first discuss Bitcoin's distributed system of governance.

3.1 The Bitcoin Improvement Proposal System

In the Internet Policy Review article *The invisible politics of Bitcoin: governance crisis of a decentralized infrastructure*, Primavera de Filippi and Benjamin Loveluck describe the Bitcoin network as being governed by “two separate, but highly interdependent layers,” these layers being the infrastructure layer, consisting of the decentralized, trustless, peer-to-peer network that operates the blockchain according to its protocols, and the architects: the small group of developers entrusted with the software design of the network.¹⁹ The infrastructure layer is primarily governed by the math that underpins Bitcoin's protocols for mining new blocks, validating transaction signatures, and

¹⁷Team, Blockstream. “Blockstream Research Demonstrates Quantum-Resistant Transaction Signing on Liquid Using Simplicity Smart Contracts.” Blockstream, April 7, 2026. <https://blog.blockstream.com/blockstream-research-demonstrates-quantum-resistant-transaction-signing-on-liquid-using-simplicity-smart-contracts/>.

¹⁸Thorn, Alex, and Karim Helmy. “Bitcoin's Taproot.” Galaxy. Accessed March 19, 2026. <https://www.galaxy.com/insights/research/bitcoin-taproot>.

¹⁹De Filippi, Primavera and Loveluck, Benjamin. 2016. “The invisible politics of Bitcoin: governance crisis of a decentralised infrastructure”. Internet Policy Review 5 (3): 10-11 DOI: 10.14763/2016.3.427. <https://policyreview.info/articles/analysis/invisible-politics-bitcoin-governance-crisis-decentralised-infrastructure>.

adjusting mining difficulty. As such, the infrastructure layer requires very little human governance to function smoothly. The architecture layer, on the other hand, requires significant governance because there is no mathematical function that can intelligently redesign and reimplement Bitcoin's software in accordance with the will of Bitcoin's global userbase. Changes to the network require Bitcoin Improvement Proposals to be created, debated, and ultimately selected. Since most Bitcoin users do not know enough about cryptography and programming to participate in the technical parts of this process, a small number of core developers have naturally become responsible for the long-term sustainability of the Bitcoin Core project.²⁰ The BIP selection process illustrates the centrality of the architectural layer by revealing a critical bottleneck to community participation. For a BIP to be considered, the source code and documentation must be available to anyone so that the global Bitcoin community has a chance to examine and debate the change. However, the official Bitcoin Core software is hosted in a GitHub repository that, while available for anyone to view and download, limits the ability to merge proposals and commit software updates to only a few core maintainers. This is a necessity because it would be impossible to operate a project of this scale and importance with software that anyone on Earth is allowed to tamper with. Consequently, the actual selection and implementation of a BIP relies on developers assessing the degree of public support for it and coming to a consensus amongst themselves prior to merging the BIP with the Bitcoin Core repository.²¹ Given that there is no mechanism to gauge whether the majority of Bitcoin users support a BIP, this is ultimately a judgement call, which critics argue is not particularly democratic. As a counterpoint to this criticism, however, there is a form of soft "veto power" that can be exercised by the network at the architectural layer.

After merging a BIP with Bitcoin's official software package, the developers face the trickiest step in upgrading the blockchain. They must acquire the consent of the miners and node operators, as they cannot force a decentralized, anonymous, global community to simultaneously upgrade the software that is running on their machine(s). Filippi and Loveluck describe this as "market-based governance," which acts as a de-facto poll to determine which version of the Bitcoin software is preferred by the community.²² The process often involves miner signaling, which is a mechanism through which miners can broadcast that they accept, and are ready to implement a change. Essentially, it works as follows: When a change to the Bitcoin software is proposed, it is assigned a specific bit in the "version" field of a block header. When a miner wants to signal support for a proposal, they simply set that specific bit to '1' in the blocks that they mine.²³ This acts as a poll. If, after a defined window (say, 1,000 blocks, for example), a majority or supermajority of

²⁰Ibid.

²¹De Filippi and Loveluck, "The invisible politics of Bitcoin: governance crisis of a decentralised infrastructure", 13-14

²²Ibid.

²³Version bits FAQ for miners was published on June 08, 2016. "Version Bits FAQ for Miners." Bitcoin Core. Accessed March 19, 2026. <https://bitcoincore.org/en/2016/06/08/version-bits-miners-faq/>.

those mined blocks have the signal bit set, then the proposal enters the “lock-in” state, which signals to node operators that accept the upgrade that they should begin using the new software. During the lock-in window, node operators have until the activation block specified in the new software (for Taproot that block was 709,632) to update their software. If, after that activation block checkpoint is hit, most node operators have switched to the newest software, then the rules of that BIP become active and are accepted as a permanent part of the official Bitcoin network.

By illustrating this process, Filippi and Loveluck reveal their central insight, which is that a technocratic form of governance is a consequence of Bitcoin's decentralized architecture, as it is implicitly assumed that Bitcoin's core developers know what is best for the network by virtue of their technical expertise. New upgrade proposals then only truly enter the global debate stage during the activation phase, rather than the design and technical implementation phase. They go on to argue that this centralizing design pressure has, in the past, made the proposal process contentious, as was the case during the Block Size Wars, which were a series of debates that took place between 2015 and 2017 surrounding technical proposals to increase the size of Bitcoin blocks in order to increase transaction throughput capacity.²⁴ As it became evident that these debates were rooted in philosophical disagreements about Bitcoin's protocol and future, productive debate gave way to the censorship of Bitcoin forums, personal attacks, and even Distributed Denial of Service (DDoS) attacks on node operators.²⁵ When examining the history of Bitcoin upgrades, it is hardly controversial to argue that the ease of these rollouts fall on a spectrum, ranging from highly contentious to relatively smooth. Having failed to be accepted by the majority of Bitcoin node operators, thus resulting in the creation of a rival hard fork (Bitcoin Cash), the proposals that sparked the Block Size Wars lie on the highly contested, high friction side of this spectrum. Fortunately, however, many would argue that the Taproot upgrade, while not entirely free of controversy, falls on the opposite end.

3.2 Tracing The Taproot Upgrade

From its formal introduction in January 2018, Taproot differed from Block Size War proposals like Bitcoin XT and Bitcoin Classic, and SegWit, which is the successful 2017 soft fork that helped resolve the block size debates by changing how block size is calculated to increase throughput without vastly increasing the data size of blocks.²⁶ This is because, from its inception, Taproot began with a measure of wide support not enjoyed by these other proposals. Schnorr Signatures promised to provide better privacy, signature aggregation and more efficient multi-signature transactions, and

²⁴“What Was the Blocksize War?: Bitstamp by Robinhood.” Bitstamp by Robinhood Trusted Crypto Exchange. Accessed April 15, 2026. <https://www.bitstamp.net/learn/crypto-101/what-was-the-blocksize-war/>.

²⁵De Filippi and Loveluck, “The invisible politics of Bitcoin: governance crisis of a decentralised infrastructure”, 16-17

²⁶“What Was the Blocksize War?: Bitstamp by Robinhood.” Bitstamp by Robinhood Trusted Crypto Exchange

was thus supported by miners, node operators, users, and developers alike.²⁷ By October 2020, the three BIPs that comprised the Taproot upgrade (340, 341, 342) had all been merged into the official Bitcoin Core software repository. By January 2021, the software had been officially released in Bitcoin Core 0.21.1.²⁸ In order for Taproot to be officially enabled, however, the Bitcoin community still needed to secure enough miner support to activate it. During the 2017 SegWit upgrade, there was considerable friction caused by miners choosing not to signal due to the incompatibility of the SegWit upgrade with a mining optimization technique called “ASICBOOST,” and a desire for larger blocks.²⁹³⁰ In an attempt to combat this, Bitcoin developers changed activation mechanisms from the scheme implemented in BIP 9 to the scheme implemented in BIP 8.

The only difference between BIP 8 and BIP 9's software is an additional parameter present in BIP 8 that is not present in BIP 9. This parameter was, and arguably still is, very controversial. The “LockinOnTimeout,” or LOT parameter defines what should occur if a majority of miners do not signal that they are ready to upgrade by the end of the specified voting window. If LOT is set to false, then the upgrade simply will not go through. The “lock in” period does not begin after the time out, and developers are forced to re-evaluate and try again with another voting window. If the parameter is set to true, however, the lock in period begins, and node operators are encouraged to begin upgrading their software. Miners are then forced to upgrade their software too, because if they don't, nodes will begin rejecting “outdated” blocks that are mined after the activation block. Once a change was proposed, the decision of whether to set the LOT parameter to true or false immediately erupted. Some argued that it was best to give miners the ability to reject a proposal so as to ensure that the entire Bitcoin community is truly on board with a change. Others argued that this gave miners too much power in vetoing changes that were universally agreed upon as good for the network. The debate became less about Taproot itself, and mostly about what mechanisms should be used to implement a change, regardless of whether that change is popular.³¹

In the spring of 2021, a compromise was proposed by Bitcoin developers David Harding and Russell O'Connor that came to be known as the “Speedy Trial.”³² It sought to resolve the debate

²⁷Thorn, Alex, and Karim Helmy. “Bitcoin's Taproot.” Galaxy. Accessed March 19, 2026. <https://www.galaxy.com/insights/research/bitcoin-taproot>.

²⁸Ibid.

²⁹Many miners wanted larger blocks because it was believed that if it was cheap and fast to transact on the Bitcoin network, it would attract users, which would thus increase the total amount of transaction fees that could be collected. They believed this would exceed the value decrease of individual fees that would result from less competition for block space.

³⁰Torpey, Kyle. “Is a Mining Manufacturer Blocking SegWit to Benefit from ASICBOOST?” Bitcoin Magazine, April 10, 2017. <https://bitcoinmagazine.com/business/mining-manufacturer-blocking-segwit-benefit-asicboost>.

³¹Thorn, Alex, and Karim Helmy. “Bitcoin's Taproot.” Galaxy. Accessed March 19, 2026. <https://www.galaxy.com/insights/research/bitcoin-taproot>.

³²Colin Harper, Blockspace Media. “Taproot Speedy Trial Code Merged into Bitcoin Core.” CoinDesk, April 15, 2021. <https://www.coindesk.com/tech/2021/04/15/taproot-speedy-trial-code-merged-into-bitcoin-core>.

by condensing each voting window (also known as an epoch) to 2016 blocks (which amounts to roughly 2 weeks for the Bitcoin blockchain) and assigning six of them back-to-back. If, during any of these six epochs, 90% or more of the blocks mined signaled support for Taproot, the proposal would enter the lock in period. If, by August 11th, 2021, that didn't occur, the proposal would temporarily fail, and the Bitcoin community would look for alternative activation mechanisms. The highly condensed, successive voting windows enacted in this compromise appealed to much of the community, as it wouldn't automatically force miners to accept Taproot, but it would leave enough time for the users to force a soft fork if they eventually needed to.³³ This contingency wouldn't need to be explored, however, because on June 11th, 2021 the Taproot upgrade entered the "lock in" phase with over 99% hash rate approval. On Sunday, November 14th, Taproot became officially activated on the Bitcoin network.

Compared to its predecessor upgrades, the relative ease of Taproot's pre-activation debate phase is reinforced by analysis performed in the research paper *Identifying Influence Mechanisms in Permissionless Blockchain Communities: The Bitcoin Case* by Rewat Thapa, Pankajeshwara Sharma, Joschka Andreas Hüllmann and Bastin Tony Roy Savarimuthu. In this paper, the authors performed a Social Network Analysis on the Bitcoin developer mailing list during Taproot's proposal period, treating each member as a node. The edges connecting each node thus represented a recorded communication between members via the mailing list. They then calculated which members acted most as brokers by facilitating a conversation between other members, and which members had the most connections to the highest connected members.³⁴ What they found by performing further analysis on the content of the most well-connected developers' emails was that they utilized "Shepherding the community" and a "participative approach" as the primary means to win developers over to Taproot.³⁵ Both of these are collaborative and reciprocal approaches that emphasize personal persuasion and alliance building.³⁶ This research supports the notion that a smooth upgrade is not just the product of an uncontroversial proposal, but also an environment wherein well-reasoned debate and compromise is possible among members of the Bitcoin community. As we will see in the next section, these underlying assets are not necessarily a guarantee for Bitcoin's impending post-quantum transition.

³³Ibid.

³⁴Thapa, R., Sharma, P., Hüllmann, J. A., & Savarimuthu, B. T. R. (2021). Identifying Influence Mechanisms in Permissionless Blockchain Communities: The Bitcoin Case. In *Proceedings of the 42nd International Conference on Information Systems (ICIS)* (pp. 1-17). Article 1689. <https://www.joschka-huellmann.net/publications/2021-09-29-final-submission-icis.pdf>

³⁵Thapa, R., Sharma, P., Hüllmann, J. A., & Savarimuthu, B. T. R. (2021). Identifying Influence Mechanisms in Permissionless Blockchain Communities: The Bitcoin Case: 12-17

³⁶Ibid.

4. How A Post-Quantum Migration Diverges from Taproot

As discussed in the previous section, Taproot succeeded because all members of the architectural and infrastructural governing layers could be reached through some form of lateral persuasion or included in open consensus-building debates. Furthermore, the uncontroversial nature of the upgrade helped these discussions come to a productive resolution, even when there was disagreement about the activation mechanism. An examination of the circumstances surrounding the impending post-quantum transition, however, suggests that it will not share these same qualities with the Taproot debate. As Bitcoin gains mainstream acceptance, regulated institutions have become increasingly integrated into the global community of bitcoin holders. The approval of spot Bitcoin ETF's in the United States in January 2024 has accelerated this integration to a point where, as of April 2026, Bitcoin spot ETF's hold approximately \$95 Billion in total assets under management (AUM).³⁷ We see a similar trajectory for other regulated institutions like bitcoin custodians and crypto exchanges.³⁸ Firms like Coinbase, Fidelity Digital Assets, BitGo, Gemini, and NYDIG collectively custody a significant share of the bitcoin supply on behalf of their clients.³⁹ As we will see, this shift in Bitcoin custodianship places the network in a notably different position than the one it held during the Taproot upgrade, and SegWit before that. During our discussion of Bitcoin governance and Taproot, we have largely omitted any mention of bitcoin users that are not Bitcoin developers, and do not operate nodes or miners, but simply hold and use bitcoin for spending and saving. Those involved in Bitcoin's governance typically assume that these users will accept almost any change to the network so long as it does not negatively affect the operation or integrity of the network, which should be guaranteed by the fact that the Bitcoin developers, nodes and miners are also bitcoin users, and all agreed to roll out the upgrade in the first place. If they roll out an upgrade that reduces usability for the average user, they will leave and the value of the developers', node operators' and miners' holdings will crash. As we will see, however, this assumption of implicit acceptance does not necessarily hold if the bitcoin user is, for example, a regulated financial institution holding massive amounts of bitcoin in cold storage to back a bitcoin spot ETF. This is a critical divergence for Bitcoin's post-quantum migration.

³⁷"Bitcoin ETF Fund Flows: Spot BTC Net Inflow & Holdings." coinglass. Accessed April 15, 2026. <https://www.coinglass.com/etf/bitcoin>.

³⁸Braun, Helene. "Crypto News: Survey Finds Institutional Investors Planning to Boost Allocations." Crypto news: survey finds institutional investors planning to boost allocations, March 18, 2026. <https://www.coindesk.com/business/2026/03/18/large-investors-are-doubling-down-on-crypto-but-getting-a-lot-pickier-about-risk>.

³⁹Okx. "BTC and Fidelity: How Institutional Custody Is Shaping the Future of Bitcoin." OKX United States. Accessed April 15, 2026. <https://www.okx.com/en-us/learn/btc-fidelity-institutional-bitcoin-custody>.

4.1 Financial Institutions as an Emerging Party in Bitcoin Governance

For much of the world, private, self-custodial bitcoin holders do not face easily enforceable restrictions or regulations that dictate how they can privately hold cryptocurrency, as any law attempting to regulate key generation and storage on private computers would be difficult to implement. This is not the case for large custodial institutions and the enterprise-grade hardware security modules (HSMs) they use to store cryptographic keys, which can be easily visited and audited by a public or private standards agency. In their paper *Crypto Custody: An Empirical Assessment*, Dirk Zetsche and Areti Nikolakopoulou examine the terms and conditions set by leading crypto custodians. Their findings support the assertion that there is an ongoing “intermediation and institutionalization” of crypto that necessarily requires a “sophisticated crypto custody framework.”⁴⁰ The authors also claim that, as such, we should expect an increase in financial institutions and standards bodies influencing custodial practices.⁴¹ A perfect example of this, and one where the technical implementation of the blockchain is relevant, is the European Union’s 2023 Markets in Crypto-Asset (MiCA) Regulation. MiCA is a set of rules and regulations that must be followed by European institutions that handle crypto assets. An example of one type of regulation included is the requirement that crypto custodians must disclose custody policies, like whether keys are being held in hot and cold storage, to clients. This framework also permits custodians to utilize omnibus accounts, which are wallet addresses that store the assets of multiple clients. The authors note that the MiCA rules surrounding the safekeeping of crypto assets link “secure crypto custody” to the protection of clients’ rights, and more importantly, to the prevention of the rehypothecation or reuse of their crypto assets.⁴² While it might initially seem great that these requirements exist, this regulation could have very severe implications during a post-quantum upgrade. Later in their paper, Zetsche and Nikolakopoulou elaborate on “Rights of Reuse” being applied to crypto assets. In a traditional financial system, they argue, the reuse of an asset in custody refers to the “transferring, pledging, selling, lending, or investing” of that asset.⁴³

With this regulatory prohibition on transferring clients’ crypto assets between wallets in mind, consider a recent post-quantum BIP, which is currently being debated. On July 12th, 2025, Jameson Lopp, a prominent Bitcoin developer published a proposal on the bitcoin developer mailing list titled “A Post Quantum Migration Proposal.” It is an aggressive proposal that looks to engineer urgency in order to force users to migrate their funds from quantum-vulnerable wallets to quantum

⁴⁰Dirk Zetsche, Areti Nikolakopoulou, *Crypto Custody: An Empirical Assessment*, *Journal of Financial Regulation*, Volume 11, Issue 1, April 2025, Pages 74–75, <https://doi.org/10.1093/jfr/fjaf004>

⁴¹Dirk Zetsche, Areti Nikolakopoulou, *Crypto Custody: An Empirical Assessment*, *Journal of Financial Regulation*, Volume 11, Issue 1, April 2025, Pages 83–86, <https://doi.org/10.1093/jfr/fjaf004>

⁴²Ibid.

⁴³Dirk Zetsche, Areti Nikolakopoulou, *Crypto Custody: An Empirical Assessment*, *Journal of Financial Regulation*:

secure ones.⁴⁴ It would do this in three phases. Phase A would begin three years after the activation of BIP 360 and would revoke the ability to send funds to a legacy ECDSA/Schnorr address.⁴⁵ This would mean that users could only send bitcoin to a quantum-secure P2QRH wallet.⁴⁶ Phase B of this plan would begin two years after Phase A was enacted and would prevent the spending of any remaining UTXO from a quantum-vulnerable account.⁴⁷ This means that unspent, unmigrated bitcoin would become unusable by the owner. Phase C is more of a hypothetical suggestion, allowing users to reclaim frozen bitcoin by constructing a quantum-safe ZK proof with the corresponding seed phrase. There is no timeline for phase C, however, because it would require a hard fork of the blockchain and hasn't been fully researched yet.⁴⁸ While this proposal is still in the early stages of debate, it represents perhaps the most coherent high-level migration proposal actively being debated by Bitcoin developers.

If this BIP were enacted while custodial institutions are unable to legally transfer funds from legacy omnibus wallets to quantum-secure omnibus wallets without the consent of all their clients, this could cause very severe friction. It is likely that proactive or high-net worth individuals would attempt to pull their crypto out of the custodian's wallet so that they could personally and safely transfer it to a quantum-secure wallet without having to risk waiting for every other client to give their consent. This bank run-like cascade could be ruinous for that institution, as many crypto custodians generate revenue by charging a fee to store a client's bitcoin. As such, it follows that large crypto custodians are incentivized to oppose this aggressive BIP, or any BIP that would make it difficult or impossible to remain compliant with the regulatory framework they are bound to. Thus, financial institutions may attempt to exert influence over Bitcoin governance in a fundamentally new way. In our earlier examination, we identified four primary groups that are relevant to the governance of Bitcoin: miners, nodes, developers and ordinary bitcoin users (that do not mine or run a node). We now see that financial institutions are emerging as a relevant fifth party. These financial institutions cannot be reasoned with or persuaded of a BIP's efficacy in the same way that the original four can because you simply cannot convince a financial institution that a new protocol change is a good plan if they cannot legally abide by that protocol change. Furthermore, unlike two developers on the same mailing list, it is more challenging to have prolonged discourse with an institution. A single developer can read a message in a thread, think about it, and update their position in an afternoon. Institutions, on the other hand, are typically represented by a team that

⁴⁴Lopp, Jameson. "A Post Quantum Migration Proposal." Google. Accessed March 18, 2026. <https://groups.google.com/g/bitcoindex/c/uEaf4bj07rE/m/wyzDA6tdBQAJ>.

⁴⁵Ibid.

⁴⁶Note that during July 2025, BIP 360 was a work-in-progress, and included embedded post-quantum signature algorithms, calling the output type Pay-to-Quantum-Resistant-Hash" (P2QRH). It was later stripped back to become the P2MR that we now associate with the BIP 360 draft that was merged with the official repo.

⁴⁷Ibid.

⁴⁸Ibid.

would have to discuss a suggestion or proposal, come to a consensus, formulate a response, and ensure that it complies with the technical and legal standards they must abide by. This, naturally, takes much more time, and is far more opaque to the party communicating with that institution. This standards-induced friction sets a potential post-quantum transition apart from Taproot's first successful attribute: productive and cooperative governance. But what about Taproot's second attribute: an uncontroversial change to the protocol?

4.2 Geopolitical Disagreement over Post-Quantum Cryptography

Unlike Schnorr signatures, which the Bitcoin community broadly agreed were a technical improvement, post-quantum signature schemes do not offer a clean, uncontroversial consensus. The selection of a post-quantum signature scheme for the Bitcoin network is set against the backdrop of a fractured political world, where different governments are establishing unique standards and requirements across a variety of competing signature schemes. In July 2025, the Post Quantum Cryptography Coalition, a group of researchers, technologists and cryptographic experts put together a report on International PQC requirements. What they found was that even among Western allies, there is considerable disagreement. The Internet Engineering Task Force (IETF) has standardized the state-based hash signatures, XMSS and LMS. The United States' NIST, on the other hand, has only standardized ML-DSA, FN-DSA (pending) and SLH-DSA.⁴⁹ The U.S.'s Commercial National Security Algorithm Suite 2.0 (CNSA 2.0) recommends ML-DSA for general signature use, and LMS/XMSS only in specific situations, like firmware signing.⁵⁰ They will not allow the use of SLH-DSA, despite NIST approving it for use in systems that aren't critical to national security. Germany's BSI, (the country's federal office for information security) does recommend SLH-DSA, as well as ML-DSA. France recommends every NIST-approved signature scheme, in addition to LMS and XMSS. Spain recommends ML-DSA, SLH-DSA and XMSS, but not LMS.⁵¹ Which algorithms are approved for use by which countries is not important for this discussion, but it demonstrates that there are many signature schemes and differing opinions on which are the best.

This disagreement is exacerbated when you include countries that do not want to use signature schemes developed by the U.S. and Western European researchers at all. As Nicholas Zúñiga, Saheli Datta Burton, Filippo Blancato, and Madeline Carr document in their paper *The geopolitics of technology standards: historical context for US, EU and Chinese approaches*, China has "...developed a long term, holistic strategy for using technical standards to bolster its domestic

⁴⁹Kampanakis, Panos, Austin Bosarge, Shahram Jamshidi, Deirdre Connolly, and Marc Manzano. "International PQC Requirements." Post-Quantum Cryptography Coalition, July 14, 2025. <https://pqcc.org/international-pqc-requirements/>.

⁵⁰Ibid.

⁵¹Ibid.

industrial development while also leveraging international and domestic standard-setting practices to exert its place as a global technology power”⁵² There is an emphasis on non-Western economic groupings like BRICS, where China hopes to partner with growing economies that exist on the margins of existing technology standardization systems. In doing so, the authors argue, China is working to leverage its growing geopolitical influence to establish a new standards systems based on Chinese values.⁵³ China benefits from these systems because they promote Chinese industry, allow China to administer industrial policy, give China greater market access, the ability to ensure quality supervision, and of course, enhanced technological security.⁵⁴ Naturally, this mission extends to cryptographic algorithm suites as well. In February 2025, China's Institute of Commercial Technology Standards launched a global call for post-quantum algorithm proposals. Though none have been officially announced yet, it is believed that they will utilize a structureless lattice design, unlike NIST-approved ML-DSA and FN-DSA, which use algebraic lattices. Many Chinese researchers believe that these algebraic lattice algorithms may be broken in the future.⁵⁵ Beyond China, The Post Quantum Cryptography Coalition's report shows that South Korea has approved four of its own unique digital signature schemes: NTRU+ and SMAUG-T as KEMs and ALMer and HAETAE.⁵⁶ Russia, too, is expected to develop its own PQC algorithms as a part of their GOST classical domestic suite.

4.3 Geopolitical Rivalry and Bitcoin's Migration

A threat to the “global” aspect of Bitcoin, thus, emerges from the geopolitically fractured nature of modern post-quantum cryptographic standards. For example, FIPS 140 is a federal standard administered jointly by the United States' NIST and the Canadian Center for Cyber Security through the Cryptographic Module Validation Program (CMVP) which requires any hardware, software, or firmware performing cryptographic functions to be independently tested and certified against a set of tiered security requirements before it can be used to secure government information. If the Bitcoin network were to implement a signature scheme that was not approved by NIST, government agencies and the financial institutions that choose to tether themselves to these standards across the U.S. and

⁵²Nicholas Zúñiga, Saheli Datta Burton, Filippo Blancato, Madeline Carr, The geopolitics of technology standards: historical context for US, EU and Chinese approaches, *International Affairs*, Volume 100, Issue 4, July 2024, Page 1636, <https://doi.org/10.1093/ia/iaae124>

⁵³Nicholas Zúñiga, Saheli Datta Burton, Filippo Blancato, Madeline Carr, The geopolitics of technology standards:1649

⁵⁴Ibid.

⁵⁵Sharma, Shweta, Ashish Mishra, and Anjali Gopinadhan Nair. “Beijing Wants Its Own Quantum-Resistant Encryption Standards Rather than Adopt NIST's.” CSO Online, March 19, 2026. <https://www.csoonline.com/article/4147736/beijing-wants-its-own-quantum-resistant-encryption-standards-rather-than-adopt-nists.html>.

⁵⁶Kampanakis, Panos, Austin Bosarge, Shahram Jamshidi, Deirdre Connolly, and Marc Manzano. “International PQC Requirements.” Post-Quantum Cryptography Coalition, July 14, 2025. <https://pqcc.org/international-pqc-requirements/>.

Canada would be unable to legally sign bitcoin transactions within a FIPS-140 validated wallet or HSM. Clients with bitcoin being held custody by these firms would be unable to spend their bitcoin, and it is likely that their funds would never be moved into a quantum-safe wallet to begin with. Conversely, if the Bitcoin network embraces ML-DSA or another NIST-approved scheme, countries that are not aligned with the United States could face similar problems. In Russia, a “Qualified Electronic Signature” (QES) must comply with domestic GOST cryptographic standards and be issued by hardware certified by a certification center accredited by Russia’s Federal Security Service (FSB).⁵⁷ This QES standard is required for financial transactions, taxes, and high-stakes commercial activity under Russia’s Federal Law No. 63-FZ.⁵⁸ Furthermore, as of March 2026, the Russian State has approved draft regulation that states that all domestic crypto trades must flow through licensed Russian intermediaries.⁵⁹ If Bitcoin does not select a Russian state-designed post-quantum signature scheme (which they almost certainly will not), then these licensed intermediaries might be unable to support Bitcoin while remaining in full compliance with domestic law. These, of course, are just a possibility for now, but they begin to illustrate a few of the ways that the national security laws of rival powers could create problems that will exert friction in the proposal phase of Bitcoin’s post-quantum migration.

4.4 The Cryptographic Politics of Strategic Bitcoin Reserves

An analysis of how states might react to Bitcoin’s post-quantum migration is made more interesting by the fact that several countries now have official, or unofficial, strategic bitcoin reserves. As of the writing of this paper, The United States and El Salvador have officially dedicated strategic bitcoin reserves, and Pakistan has announced the creation of one. The United Kingdom, Ukraine, Bhutan, Germany (and likely many others, though these holdings aren’t always disclosed) have some form of state-owned holdings.⁶⁰ China is an interesting case, because although they have banned the use of bitcoin and do not publicly disclose their holdings, it is believed that at one point they held roughly 190,000 BTC which they confiscated taking down the PlusToken Ponzi scheme in 2019.⁶¹ For a time, this made China the second largest state holder of bitcoin (after the United

⁵⁷ Baydakova, Anna. “Russia’s FSB Is Making Life Harder for Blockchain Companies.” Yahoo! Finance. Accessed April 16, 2026. <https://finance.yahoo.com/news/russia-fsb-making-life-harder-161511758.html>.

⁵⁸ Shunfang. “Russia Federal Law No 63-FZ: Electronic Signatures Guide.” eSignGlobal, 2026. <https://www.esignglobal.com/blog/russia-federal-law-63-fz-electronic-signature>.

⁵⁹ ForkLog. “The End of Shadow Trading? Russia’s Forthcoming Crypto-Market Rules: Forklog.” forklog.media, March 23, 2026. <https://forklog.com/en/the-end-of-shadow-trading-russias-forthcoming-crypto-market-rules/>.

⁶⁰ Bleep. “Cryptocurrency Reserve by Country (2026): Who Has a Crypto National Reserve and Who Holds the Largest?” Bleep. Accessed April 17, 2026. <https://www.bleep.finance/en-us/blog/cryptocurrency-reserve-by-country>.

⁶¹ Prashant, Jha. “Institutions Selling Bitcoin near \$70k- Do They Know Something the Market Doesn’t?” Yahoo! Finance. Accessed April 17, 2026. <https://finance.yahoo.com/markets/crypto/articles/institutions-selling-bitcoin-near-70k-092025785.html>.

States which holds roughly 328,000 BTC). Though evidence suggests that China has sold much of this stash already, it is likely that they still hold some bitcoin in accounts not publicly known to be handled by the Chinese state. The existence of these reserves raises the security stakes for this post-quantum migration considerably, because the algorithm choice now affects a national asset. Regardless of how well-studied and transparent the development process of a cryptographic algorithm is, if you are a state like Russia or China, would you feel comfortable allowing your reserves of an asset that you consider to be strategically important to be rebuilt according to NIST, a standards body heavily affiliated with the NSA? Alternatively, if you are the United States, now the largest state owner of Bitcoin, is it not in your best interest to ensure that the protocol is governed by standards designed and published by your own agencies? These questions might seem far-fetched, but it is broadly acknowledged that the U.S. and China both understand the strategic importance of influencing global trade and monetary networks, especially by designing or facilitating underlying network technologies. This is evident by China's attempts to lessen dependence on the U.S. dollar as a reserve currency, and to develop Chinese-led alternatives to U.S.-aligned financial systems like SWIFT, such as the Cross-Border Interbank Payments System (CIPS).⁶² Under these circumstances, it is not difficult to imagine a world in which a team of NSA cryptographers might be tasked with exerting pressure, either publicly or covertly, on bitcoin developers during the technical development phase in an attempt to push them towards a NIST-aligned signature scheme. This, or any similar form of national persuasion, is another plausible form of external geopolitical pressure that could complicate discussions about Bitcoin's post-quantum transition, further differentiating this upgrade process from Taproot's.

Altogether, we have identified two major divergences from the Taproot upgrade rollout. They are, first, the emergence of regulated institutions as a fifth governance constituency, and second, the geopolitically fragmented nature of the cryptographic upgrade in question. This suggests that the Bitcoin community will face a fundamentally different upgrade process than with Taproot, with its own unique challenges. Taproot's primary hurdle was the result of internal conflict between voluntary operators of the network and was resolved through the "Speedy Trial" creative solution. Bitcoin's impending post-quantum transition will face increasing external friction from users bound by regulatory frameworks, which Bitcoin's current governance model has no mechanism to address, and from geopolitical rivalries, which have no interest or reason to accommodate Bitcoin's cryptographic sovereignty. This leads to a critical question about Bitcoin's future: Even if the Bitcoin network can preserve the integrity of the blockchain against the rise of quantum computing, will it be able to preserve its status as a useful, cross-jurisdictional monetary network?

⁶²Eichengreen, Barry. "Sanctions, Swift, and China's Cross-Border Interbank Payments System." CSIS. Accessed April 17, 2026. <https://www.csis.org/analysis/sanctions-swift-and-chinas-cross-border-interbank-payments-system>.

5. Post-Quantum Migration and the Future of Bitcoin

Codified in Satoshi Nakamoto's 2008 Bitcoin white paper, the Bitcoin network was designed to be a politically neutral, peer-to-peer digital cash system. It rests atop a trustless, borderless network that does not require the resources or permission of financial institutions or trusted third parties. Early on, this decentralized neutrality was fairly easy to maintain, as the project was small and most of Bitcoin's users were individuals operating entirely outside traditional financial institutions. For better or for worse, this is no longer the case. Custodial platforms managing billions in crypto assets have brought Bitcoin into the regulatory fold via new frameworks that govern these platforms like traditional financial institutions. It is this mainstreaming of Bitcoin that will be working against it in the next major upgrade. Following an analysis of Bitcoin's 2021 Taproot upgrade, we have seen two notable ways in which the Bitcoin network's post-quantum migration is likely to diverge from Taproot, resulting in a more challenging migration. With this information, we are now equipped to examine what the future of the Bitcoin network might look like if a post-quantum cryptographic consensus is not reached. To do so, it is useful to consider three scenarios, each representing a plausible path forward, and each carrying its own set of implications for Bitcoin's future as a truly global monetary system.

5.1 NIST-aligned Signature Scheme

The first, and perhaps most straightforward path is for the Bitcoin network to adopt a NIST-approved signature scheme. The most likely option would be a variant of ML-DSA, though perhaps an SLH-DSA scheme could be designed as a slightly less efficient but more conservative solution. This would satisfy large swathes of Bitcoin's userbase, particularly American government and financial institutions, and Western European financial institutions that follow NIST's lead with minor modifications. This would be the preferred solution of U.S. institutions because, according to interpretive letters 1170, 1172 and 1174, the U.S.'s Office of the Comptroller of the Currency has clarified that national banks and federal savings associations are allowed to custody cryptographic assets.⁶³ In doing so, however, their custodial activities are subject to the OCC's existing banking standards.⁶⁴ This means that these institutions will need to implement a custodial solution that remains compliant with NIST's Federal Information Processing Standards (FIPS) in order to continue working directly with federal agencies. This option would alleviate transitional friction in Western countries, but cost support in regions that arguably need Bitcoin the most.

⁶³"OCC-Approved Crypto Custody: What Institutions Should Know." The Digital Asset Infrastructure Company, December 23, 2025. <https://www.bitgo.com/resources/blog/occ-approved-crypto-custody/>.

⁶⁴Ibid.

Following a NIST-led migration, countries that do not align with the U.S. or have had their economies and currencies crippled by sanctions might impose restrictions that make it harder to access the Bitcoin network. Countries with authoritarian governments might use it as an excuse to limit or outright ban the use of bitcoin for being a Western-dominated monetary tool meant to undermine their financial authority. In the case of China, for example, a NIST-driven migration might vindicate their ban on the use of Bitcoin, further harming bitcoin's prospects of ever becoming a truly global currency by disincentivizing China from embracing a more liberal position. This is because doing so would, in accordance with China's 2019 Encryption Law, require China's State Cryptography Administration to add NIST's algorithms to its approved list, and would give a supply-chain advantage to U.S.-aligned hardware vendors at the expense of China's domestic cryptographic industry.⁶⁵ If we stretch this scenario to the extremes, we could even imagine a world where Bitcoin begins to resemble traditional dollar-dominated financial structures like SWIFT, wherein Bitcoin is interoperable with Western economies, but treated as a foreign system by countries operating outside, or at the periphery of, the U.S.-led financial order. This is especially limiting to the adoption of Bitcoin as a global currency because developing nations, sanctioned economies, and regions experiencing hyperinflation have perhaps the most to gain from adopting bitcoin as a means of building long-term wealth and circumventing dollar dependency. Understandably, this adoption pitch is weakened if the blockchain is governed by U.S. government-administered cryptographic standards that were embraced to minimize friction for the same U.S. financial firms that these nations are trying to remain independent from.

5.2 Multiple Signature Schemes

The second scenario appears to be possible, if not likely, given current developments in Bitcoin's post-quantum migration. BIP 360's P2MR output type deliberately avoids selecting a post-quantum signature scheme. In addition to buying developers more time by kicking the problem down the road, this is beneficial because, as was previously mentioned, it gives developers the option to add multiple signature schemes via the Bitcoin scripting language. The appeal of this approach is immediately apparent: the Bitcoin network could claim to be cryptographically neutral by enabling the use of multiple signature scheme that vary in their geopolitical origin. When constructing transactions, users themselves would choose to utilize whichever signature schemes they wanted. American users could transact with NIST-approved signature schemes only, users in BRICS countries could use a Chinese structural lattice algorithm, European users could use a stateful-based hash signature, and so on and so forth. The problem with this system is that, while it

⁶⁵Chinalawtranslate. "Cryptography Law of the P.R.C." China Law Translate, 2020. <https://www.chinalawtranslate.com/en/cryptography-law/>.

might improve global access to the blockchain, it could also end up weakening it by diminishing the fungibility of each bitcoin.

For example, If Alice wants to send Bob some bitcoin, she signs her inputs with her key and locks it to the address corresponding to Bob's wallet. Suppose, now, that Alice is using a non-NIST approved signature scheme, and Bob's bitcoin wallet is in the custody of a FIPS compliant institution. To spend that bitcoin, the hardware security module (HSM) acting as Bob's wallet would likely attempt to validate that coin's UTXO history before notifying Bob of his increased funds. Since it passed through a non-NIST address, however, the HSM will be unable to perform the math required to verify its cryptographic history. The custodial institution can trust that it is a valid transaction based on its acceptance by the network, but it cannot validate the spend itself. While this doesn't wreck the functionality of the network, it does make it difficult or impossible for financial institutions to cleanly audit the movement of their crypto assets. It also harms the basic monetary principle of fungibility, which states that each unit of money should be identical to and replaceable by every other unit of that money. Over time, this could erode trust in the Bitcoin network, causing adoption to slide backward, away from mainstream institutions and back towards a niche, small community of supporters.

5.3 Failure to Reach a Consensus

The third scenario arguably requires the least elaboration and is almost certainly the worst for the future of the Bitcoin network. As we've seen throughout this paper, upgrading the Bitcoin network is a slow, methodical process that becomes more difficult under political pressure. The Block Size Wars lasted years and failed to result in a compromise that satisfied the entire community. The LOT debate over Taproot's activation mechanism was contentious enough that a new mechanism had to be created to resolve the conflict. If you combine a contested field of signature candidates with an increasingly complex and intractable governance system, it is very possible that the BIP process stalls altogether. If the consequences of a post-quantum migration are economically severe enough, then the pressure exerted by the closing migration window might not be enough to force a consensus. Jameson Lopp's quantum migration timeline BIP reveals that already there are Bitcoin developers that lack confidence in the ability of Bitcoin's governance structure to voluntarily migrate. The coercive phases he describes (preventing users from sending bitcoin to legacy addresses and then eventually freezing unmigrated UTXOs) seem almost infinitely more contentious than the debate over whether a single parameter should be set true or false.

To some, the freezing of unspent bitcoin would amount to a tremendous hit to their trust in the network. Bitcoin is supposed to be ungovernable money that cannot be frozen or seized by a central authority. If the network decides to freeze users' funds now, who's to say the network

couldn't do it again in the future? If Bitcoin's post-quantum migration does not involve freezing UTXOs, however, the network would be allowing whatever authority has first access to a CRQC to steal the bitcoin of every address vulnerable to a long exposure attack, stretching all the way back to, and including, Satoshi Nakamoto himself. Some bitcoin users would have their confidence in the Bitcoin project destroyed by that event. If attackers can siphon billions of dollars' worth of bitcoin out of users' legacy addresses, then it is not truly cryptographically secure. Why bother trying to store and accrue value in bitcoin over a long period of time if it is possible that, every few decades, a cryptographic breakthrough could give attackers the ability to steal all your money? It is very possible that we'd see one or more hard forks or chain splits, or users fleeing to other cryptocurrencies. This migration could fracture the Bitcoin community, ending any hope for a global Bitcoin standard. For some, seeing the Bitcoin network struggle to resolve the debates over which signature schemes to utilize and what to do with unmigrated, vulnerable UTXOs will make them reconsider using bitcoin as a long-term store of value. Maybe, it would seem, the Bitcoin network is not as secure as the cryptography it is built upon, because even the most air-tight cryptographic implementation of the protocol depends upon a nexus of fragile human connections that must work together to govern, redesign and reimplement the network in the face of novel threats.

Across all three scenarios, the institutions most at risk during a post-quantum migration are the same ones working hard to make bitcoin into a mainstream financial asset. Regulated exchanges, ETF managers and other crypto custodians bridge the Bitcoin network to the world of traditional finance, and it is this precarious position that is most exposed to the risks posed by technical changes to the underlying Bitcoin protocol. In the eyes of Bitcoin's earliest developers, it is perhaps a success that a self-custodial bitcoin user running a pruned node on an old laptop in their bedroom is far more insulated and financially secure than some of the largest, wealthiest asset managers in history.

6. Conclusion

Bitcoin was designed around the premise that mathematics, not institutions, could govern a politically neutral and globally decentralized monetary system. For over a decade, the network upheld that premise through a deliberate, voluntary governance model that utilized public consensus to remain functional through highly contested debates, ranging from the Block Size Wars in 2015, to the SegWit and Taproot upgrades in 2017 and 2021, respectively. When it comes to the impending post-quantum transition, the looming question is not whether Bitcoin's developers will be able to come up with a cryptographically sound solution. BIP 360, and Jameson Lopp's migration plan referencing a conceptual "Pay-To-Quantum-Resistant-Hash" output type suggest that we are closer to a technical solution than many realize. The question also isn't about whether a sound proposal

can win over a large enough consensus of the network to activate a proposal, even when there are some that disagree with the rollout mechanism. Taproot showed that this, too, is well within reason. Rather, the unanswered question looming over the post-quantum migration is about whether Bitcoin's governance mechanisms can handle the external, uncompromising pressure exerted by institutions bound to regulations imposed by standards bodies, and governments motivated by geopolitical rivalry.

A migration that is both successful and smooth is certainly possible, but it would require conditions that are not currently in place. We would either need to see a convergence of global post-quantum cryptography standards such that a single algorithm is deemed to be secure, efficient and uncontroversial by nearly every country on Earth, or the development of a Bitcoin signature framework that supports a wide variety of post-quantum algorithms and minimizes friction for institutional holders. Post-quantum standards are diverging even among Western allies, and countries like China appear fully engaged in their mission to develop domestic algorithm suites and standards. The multi-algorithm approach of BIP 360 is promising, but it is inherently harmful to the fungibility of bitcoin and could be disapproved of by powerful bitcoin custodians. In both cases, a cryptographically relevant quantum computer will likely exist long before either of these two problems resolve themselves. The purpose of this paper is not to predict that Bitcoin will fail within the next ten to fifteen years. Rather, it is to make the claim that a post-quantum migration will face external political pressures that far exceed those seen in previous upgrades. Whether Bitcoin's community can adapt to and overcome these pressures before a quantum computer renders Elliptic Curve Cryptography obsolete and fundamentally changes our digital world is, at this time, unknown. If the short and highly contested history of the world's first decentralized, peer-to-peer digital cash system is any indication, however, a creative solution might unexpectedly arise that changes how we think about the relationship between money, cryptography and decentralized governance in an inherently political world.

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